



ME 300: Engineering Thermodynamics

Example: Ideal Gas Equation of State: Calorific

Cycle example

Consider a three-step cycle with air as the working fluid: $\rightarrow$ ideal gas

- 1-2: isothermal expansion $\Delta T = 0$
- 2-3: isochoric heat addition $\Delta V = 0$
- 3-1: isobaric compression $\Delta P = 0$

$\rightarrow M = \text{const.}$

If I know the following states, solve for $\underline{P}$, $\underline{T}$, and $\underline{v}$ at all states in the cycle:

- $T_1 = 300 \text{ K}$
- $v_1 = 1 \text{ m}^3/\text{kg}$
- $v_2 = 2 \text{ m}^3/\text{kg}$
- $R = 287 \text{ J/kgK}$ - air

Δu

Δh

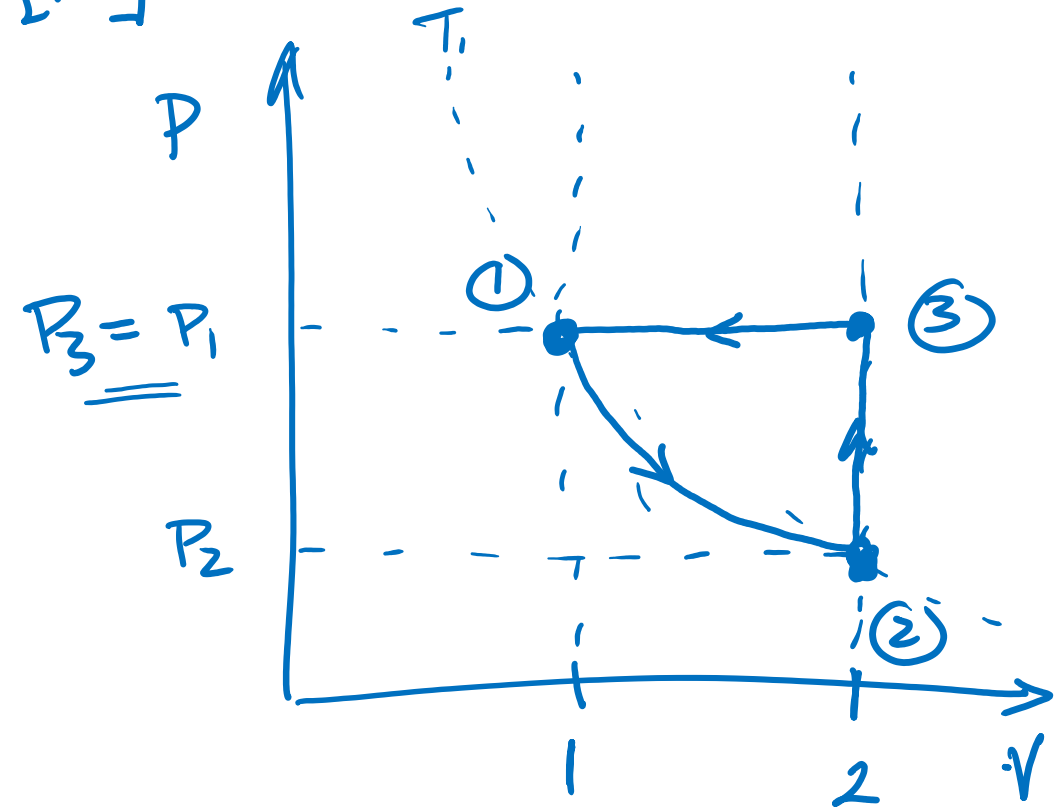
$$C_v = 714 \text{ J/kg-K}$$

$$C_p = 1001 \text{ J/kg-K}$$

Problem setup

<u>Given</u> :	<u>State</u>	$T[K]$	$v[m^3/kg]$	$P[Pa]$
	1	800	1	P
	2	800	2	$P_3 = P_1$
	3		2	P_2

Assume : ideal gas
quasi-steady.
 $C_v, C_p = \text{const.}$



Solve for states

State 1 : $T_1 = 800 \text{ K}$
 $V_1 = 1 \text{ m}^3/\text{kg}$ $\rightarrow P_1 V_1 = RT_1 \Rightarrow P_1 = \frac{RT_1}{V_1} = \boxed{229600 \text{ Pa}}$

State 2 : $T_2 = T_1 = 800 \text{ K}$
 $V_2 = 2 \text{ m}^3/\text{kg}$ $\rightarrow P_2 = \frac{RT_2}{V_2} = \boxed{114800 \text{ Pa}}$

State 3 : $V_3 = V_2 = 2 \text{ m}^3/\text{kg}$
 $P_3 = P_1 = 229600 \text{ Pa}$ $\rightarrow P_3 V_3 = RT_3 \rightarrow T_3 = \frac{P_3 V_3}{R} = \boxed{1600 \text{ K}}$

Change in energies

$$\begin{aligned} \Rightarrow \Delta u &= c_v \Delta T & \rightarrow \Delta u &= u_2 - u_1 = c_v (T_2 - T_1) \\ \Delta h &= c_p \Delta T & \Delta h &= h_2 - h_1 = c_p (T_2 - T_1) \end{aligned}$$



What is the change in internal energy for process 1-2?

- a) Greater than 0
- b) Less than 0
- c) Equal to 0

Change in energies

→ isothermal $\Delta T = 0$, $\Delta u = 0$ $\Delta h = 0$

$$\rightarrow 2-3: \underline{u_3 - u_2} = \underline{c_v} (T_3 - T_2) = 714 (1600 - 800) \rightarrow \Delta u = 571200 \frac{\text{J}}{\text{kg}}$$

$$\underline{h_3 - h_2} = \underline{c_p} (T_3 - T_2) = 1001 (1600 - 800) \rightarrow \Delta h = 800800 \frac{\text{J}}{\text{kg}}$$

$$\rightarrow 3-1: u_1 - u_3 = c_v (T_1 - T_3) = 714 (800 - 1600) = -571200 \frac{\text{J}}{\text{kg}}$$

$$\Rightarrow \sum \Delta u = 0 \rightarrow \overset{0}{\cancel{(u_2 - u_1)}} + (u_3 - u_2) + (u_1 - u_3) = 0$$
$$571200 + \underbrace{(u_1 - u_3)}_{-571200 \frac{\text{J}}{\text{kg}}} = 0$$

$$\sum \Delta h = 0 \rightarrow h_1 - h_3 = -800800 \frac{\text{J}}{\text{kg}}$$